

UCLA Stats 101C Fall 2025

Regression Project

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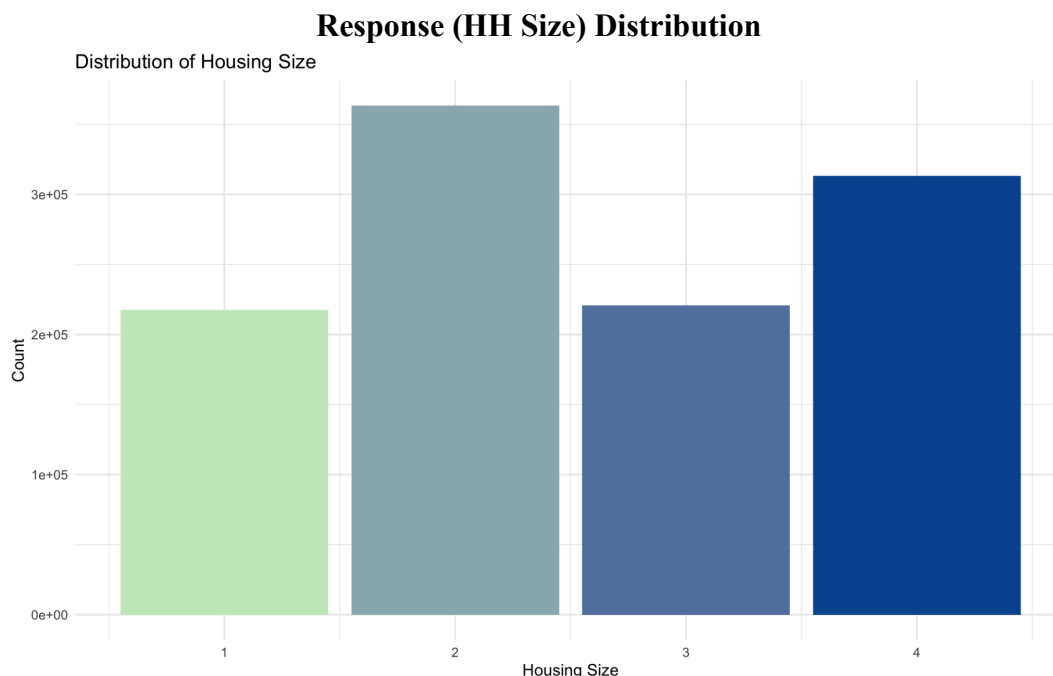
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1 Introduction

The dataset contains information about Amazon customers' purchasing histories from more than 5,000 individuals who voluntarily shared their data through Harvard's Dataverse repository. The data contains the customers' demographics, purchase categories, order frequency, and much more. Research shows that factors such as age, gender, and household composition often influence spending behavior and product choice (U.S. Bureau of Labor Statistics). Based on our purchasing experience, we expect variables such as gender, frequency of Amazon use, and spending preferences to be associated with our outcome variable: household size.

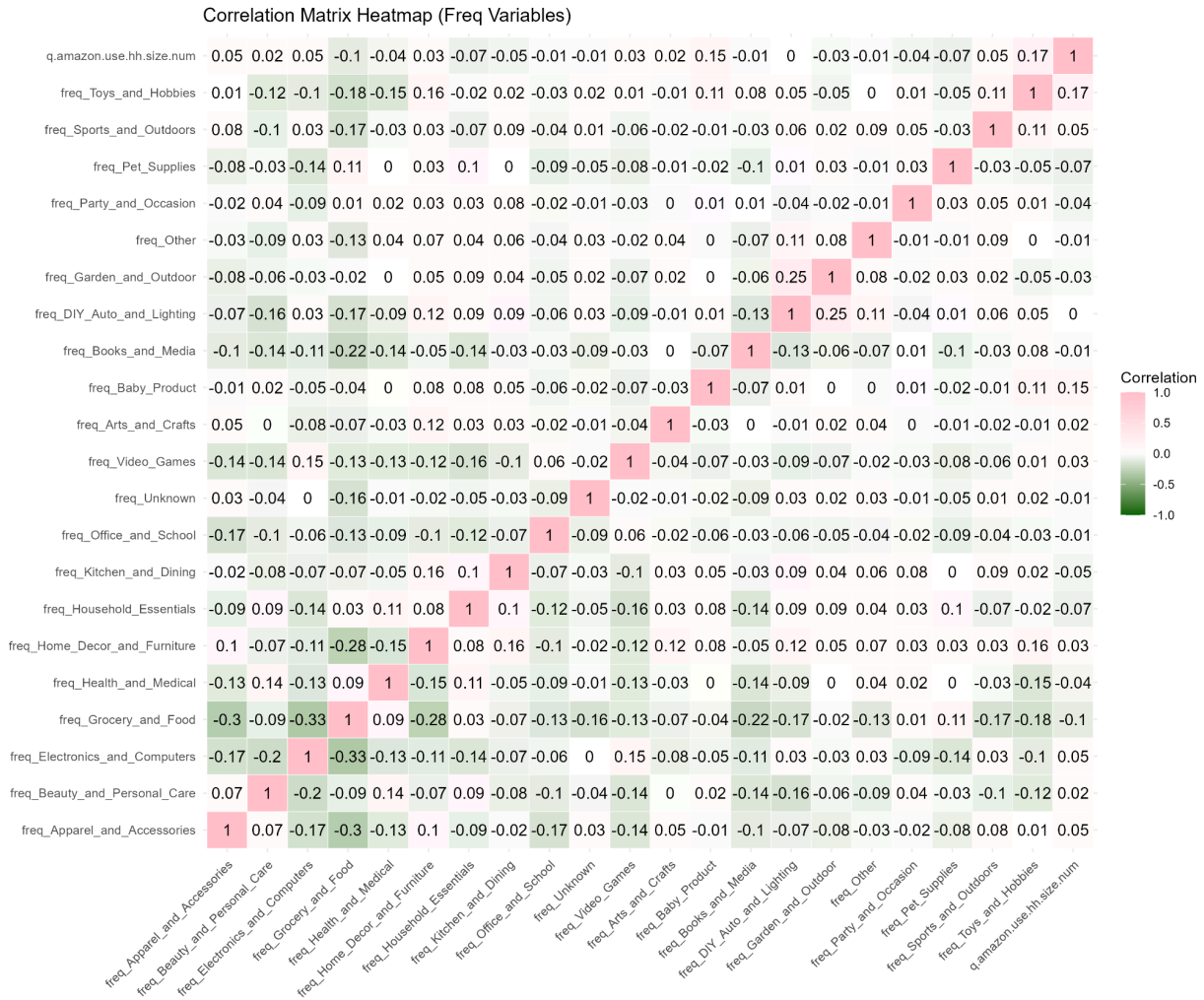
2 Exploratory Data Analysis

As we explored potential relationships between the variables in the training dataset, notable patterns were discovered, and some failed to show. Our findings are presented below:



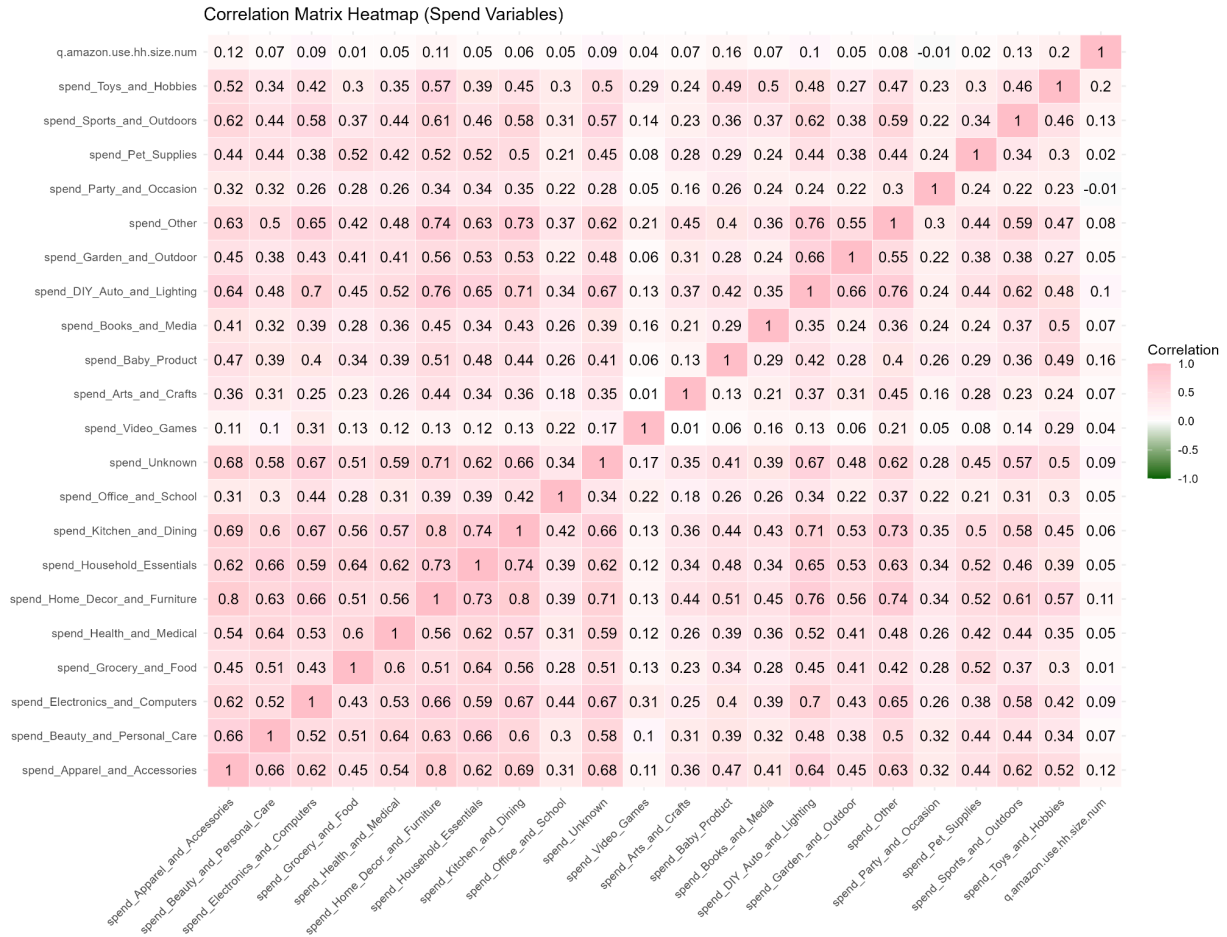
The bar chart displays the distribution of household size values in the dataset. The most common household size is 2, appearing substantially more than any other group. This suggests that a large proportion of respondents live in two-person households - likely couples living together. Household sizes of 1 and 4 also appear frequently, though not as often as size 2.

Category Purchase Frequency Correlation



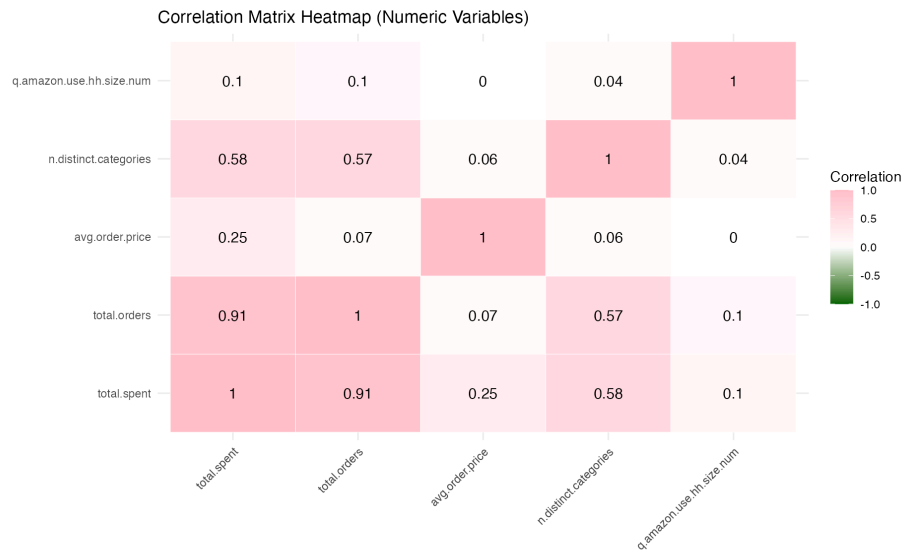
Purchasing frequency in certain categories tends to have a neutral effect on household sizing. Throughout our correlation matrix, we see clusters of slightly negative correlations amongst the different frequency categories, however, with the household size, the variables appear to have no correlation or slightly negative correlation based on the white and light green colors we see in the column and row for the household size number.

Category Total Spending Amount Correlation



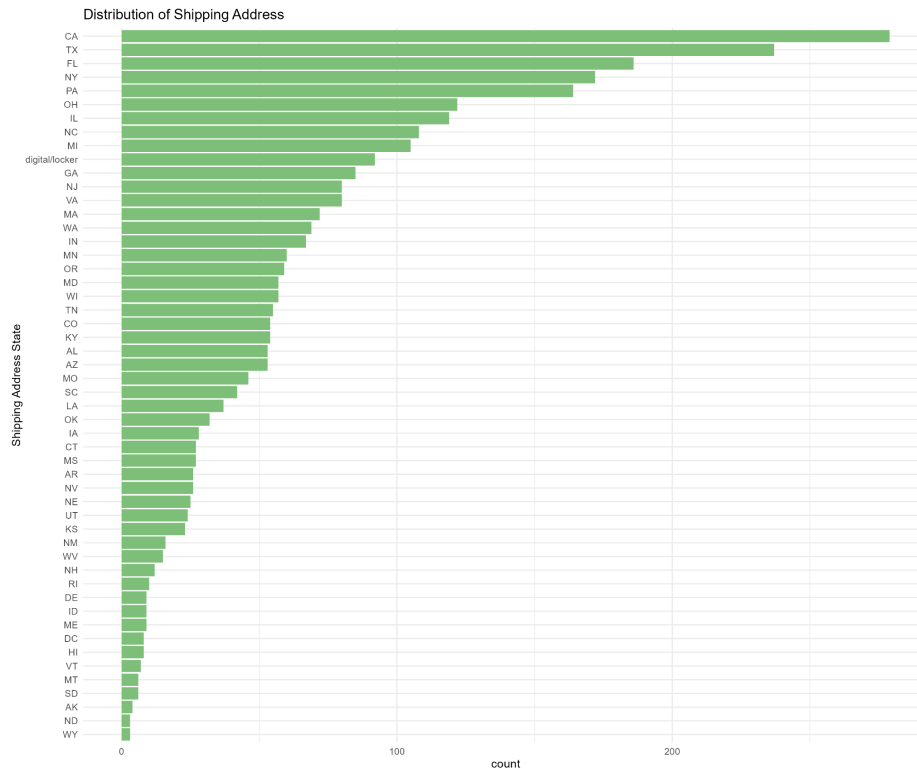
Similar to purchasing frequency, the total spending in each category also tends to have a neutral effect on household sizing. Throughout our correlation matrix, we see clusters of positive correlations amongst the different spending categories, however, with the household size, the variables appear to have no correlation or slightly negative correlation based on the white and subtle light pink colors we see in the column and row for the household size number.

Non-Category Numeric Variables Correlation

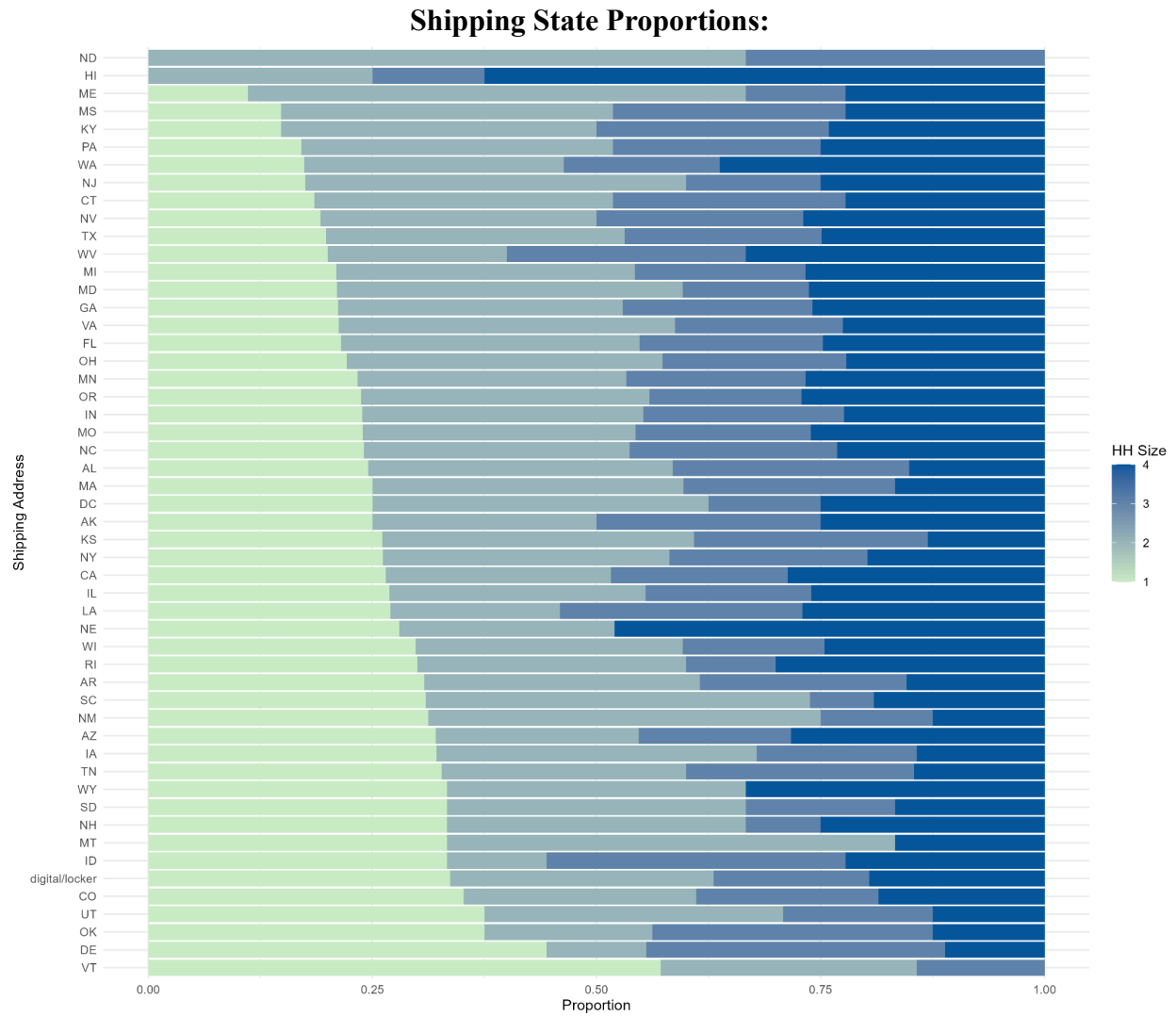


From this correlation matrix heat map of the numeric variables, we can see a few effects. The most correlated variables are those of total.orders and total.spent, with a value of 0.91, suggesting a strong positive correlation. Another relationship that we see is the moderate positive correlation between n.distinct.categories and total.spent. There are a couple variables with almost no correlation, such as total.orders, and avg.order.price.

Shipping State Distribution:



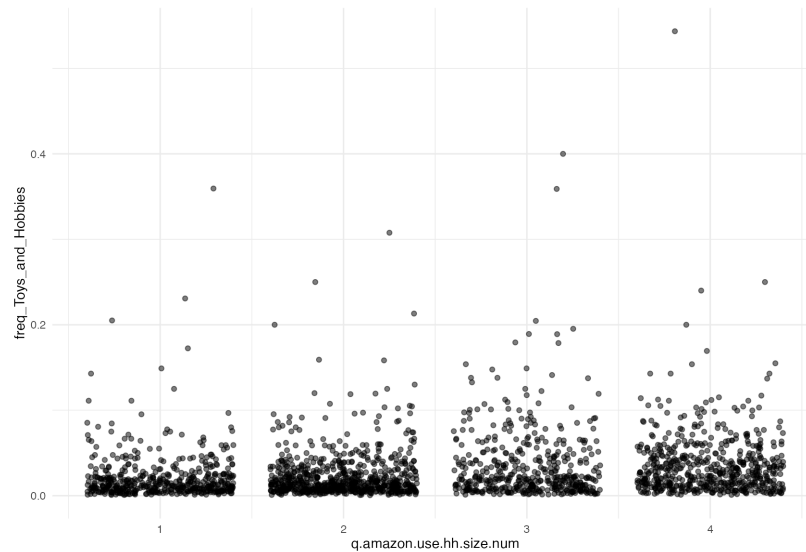
The distribution of shipping addresses that are most frequently ordered to seem to align with how populated each state is. The states we see that tend to get shipped to the most are California, Texas, Florida, and New York. These are all states that have a large population and make our visual seem pretty reasonable. Considering that some states are more populated than others, we feel that this may be a strong predictor for household size.



When combining the plot above with household sizing, we observe that high-population states (e.g., CA, TX, FL, NY) also contribute the most data in the dataset. The most common household structure across all locations is a two-person household. However, regional variations exist: Southern and Western states show slightly higher proportions of larger-size households, whereas Northern and small-population states show more 1-person households.

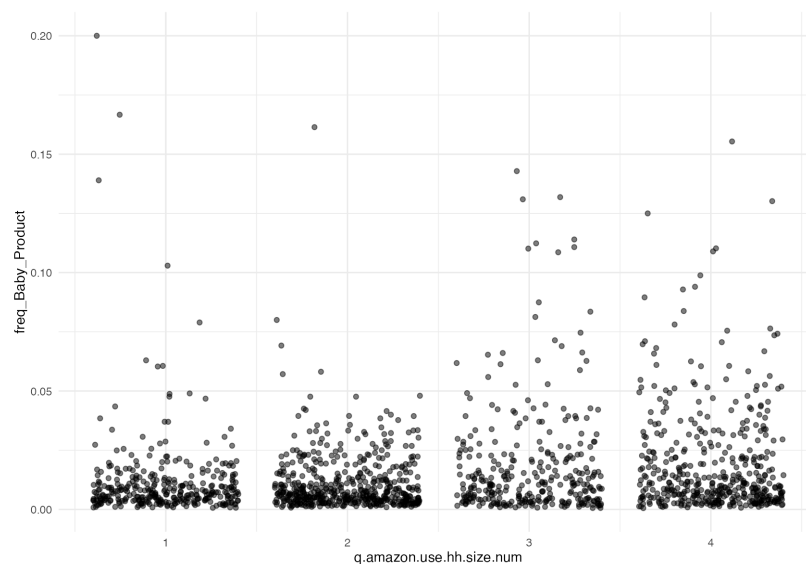
These patterns mirror real U.S. demographic trends, where households in metropolitan or suburban regions tend to have more members, while smaller states with older populations display more single-occupant households.

Household Sizes Vs. Frequency of Purchasing Toy & Hobby-Related Products



In this dot plot, we can see the frequency of toy and hobby-related purchases separated by the average household size. Surprisingly, we can see that there doesn't appear to be a large difference between each household, even though we initially predicted that having children could lead to a more frequent purchasing of toys. There are a couple outliers, particularly one household of 4, with a significant difference compared to the other households. With all the household sizes not all stratifying within a low frequency, we can assume toys or hobby-related items are not a popular item type on Amazon.

Household Sizes Vs. Frequency of Purchasing Baby-Related Products



From this dot plot, it appears that the smaller household sizes have points more densely plotted towards the lower values of the frequency scale. Many of the household sizes that have a size of

1 or 2 have lots of points within the 0.00 to 0.025 range. With our larger household sizes, we see that the points aren't as densely packed down there as the smaller households. However, we don't see the larger households with points more densely packed at a higher frequency either. Considering that this dataset captured orders from 2018 to mid-2024, larger households could have had children prior to 2018 and didn't have a need to purchase more baby-related items as the years went on.

3 Preprocessing

Within *amazon-purchases.csv*, a dataset containing user-level purchase histories, we felt that the title and product code variables were too specific and wouldn't be able to predict the household size well, so we removed them and chose to focus on the category variable instead. A major component of our preprocessing was grouping the product category information. Because the original dataset contained hundreds (~1800) of heterogeneous category labels, a grouping process was applied to reduce these into umbrella categories. Regular expressions were constructed for 19 clusters. Some examples are Electronics, Apparel, Toys, Grocery, and Household Essentials.

User features were then created. Since each customer could make multiple purchases in a day, we grouped purchases by date to represent one order. We then calculated each customer's average spending amount per order, total amount spent, and total order count. Additionally, we took the most common shipping address state per user to determine their location, and filled missing shipping address rows with "digital/locker" after looking into the initial data set's documentation (<https://www.nature.com/articles/s41597-024-03329-6#Sec10>). We also calculated purchasing trends per-user per-category, creating "freq_cat" and "spend_cat" features for each user. These outputs showed behavioral habits relevant to predicting household size.

We then began preprocessing for *survey_train_test.csv*, a dataset containing demographic and household information. Based on the project's rubric, only fields plausibly available at sign-up were retained, ensuring realistic feature availability for prediction. The only variables kept were about the user's survey ID number, gender, state, and housing size.

Afterwards, we merged the two datasets together on the user's survey ID number. Character-type variables were then converted to factors, and records containing missing target values were separated: complete cases were stored as training data, whereas records missing household size were retained as test data.

Finally, cleaned training and testing datasets were exported as *processed_data_train.RData* and *processed_data_test.RData*.

4 Candidate models / Model evaluation / Tuning

4.1 Discussion and Description of Candidate Models

The two model classes we used were Random Forest and XGboost, with different parameters resulting in a total of 5 candidate models. These 5 candidate models represent the models that have the lowest RMSE values. A brief description of each candidate model is shown below.

Table listing of all candidate models attempted:

Model identifier	Type of model	Variables in the model	Hyperparameters
tuned_lrn_xgboost1	XGBoost	• shipping.address.state • q.demos.gender • q.demos.state • total.spent • avg.order.price • freq_Apparel_and_Accessories	alpha = 0.4641589 colsample_bylevel = 0.6 colsample_bytree = 0.8 eta = 0.005994841 lambda = 46.4159 max_depth = 3 subsample = 1
lrn_rf1	Random Forest	• freq_Beauty_and_Personal_Care • freq_Electronics_and_Computers • freq_Grocery_and_Food • freq_Health_and_Medical • freq_Home_Decor_and_Furniture • freq_Household_Essentials • freq_Kitchen_and_Dining • freq_Office_and_School • freq_Unknown • freq_Video_Games • freq_Arts_and_Crafts • freq_Baby_Product • freq_Books_and_Media • freq_DIY_Auto_and_Lighting • freq_Garden_and_Outdoor • freq_Other • freq_Party_and_Occasion • freq_Pet_Supplies • freq_Sports_and_Outdoors • freq_Toys_and_Hobbies • q.amazon.use.hh.size.num	num.trees = 618 mtry = 4 min.node.size = 11 sample.fraction = 0.8449092
xgb_tune4	Random Forest	All variables above in addition to: • total.orders • n.distinct.categories	eta = 0.007055363 max_depth = 4 colsample_bytree = 0.319908 subsample = 0.369098

		● spend_Apparel_and_Accessories ● spend_Beauty_and_Personal_Care ● spend_Electronics_and_Computers ● spend_Grocery_and_Food ● spend_Health_and_Medical ● spend_Home_Decor_and_Furniture ● spend_Household_Essentials ● spend_Kitchen_and_Dining ● spend_Office_and_School ● spend_Unknown ● spend_Video_Games ● spend_Arts_and_Crafts ● spend_Baby_Product ● spend_Books_and_Media ● spend_DIY_Auto_and_Lighting ● spend_Garden_and_Outdoor ● spend_Other ● spend_Party_and_Occasion ● spend_Pet_Supplies ● spend_Sports_and_Outdoors ● spend_Toys_and_Hobbies	lambda = 0.08740375 alpha = 5.468666
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4.1.1 Model 1 (tuned_lrn_xgboost1)

This model focused on utilizing XGBoost to train an effective model. To handle categorical variables, we included a pipeline operator to one-hot encode all levels within a categorical variable. The most important hyperparameters we wanted to tune were max_depth, alpha, lambda, subsample, colsample_bytree, colsample_bylevel, and eta. To tune the hyperparameters, we created an auto-tuner with a grid search tuner to auto-tune our learner.

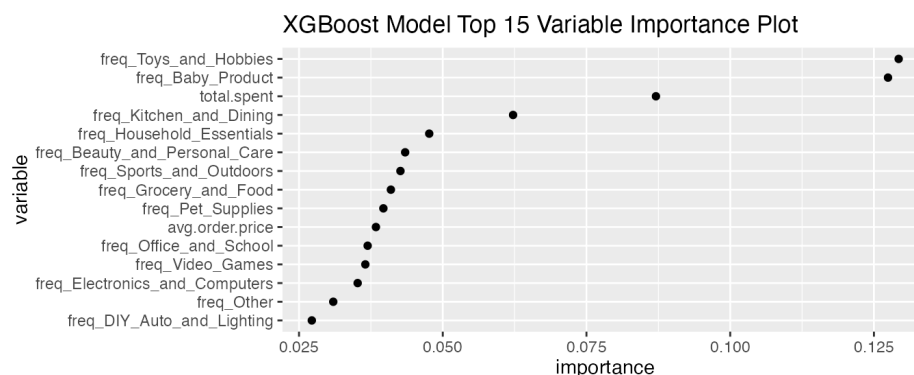


Figure 1: XGBoost Model 1- Variable Importance Plot

Based on Figure 1, it was found that how often users purchased products from specific categories and the total amount of money they spent over time explained much of the variance in our data. The predictors with the highest importance scores counted the frequency of 'Toys and Hobbies' purchases made, the frequency of 'Baby Product' purchases made, and the total amount of money spent throughout all orders by each user. This makes sense since toy and baby-related items indicate that the user most likely has a household size greater than one, and with a larger total amount of spending, having more household members to make purchases for could explain this need to spend money.

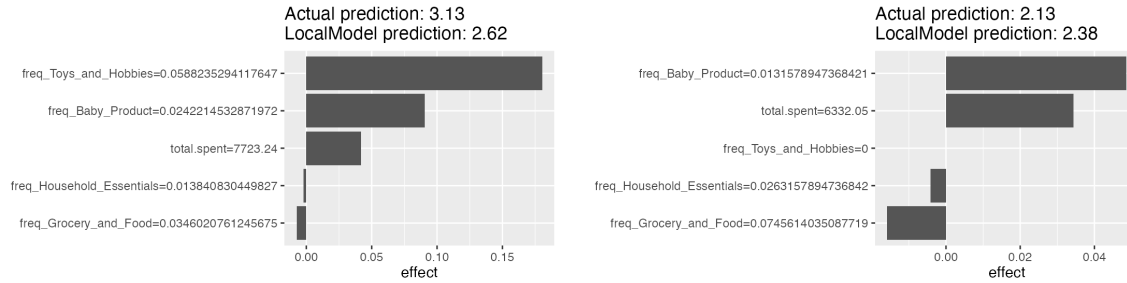


Figure 2: Left – Observation 1, Right – Observation 20

For observation 1, the actual predicted household size from the XGBoost model is approximately 3.13, while the local surrogate model approximation gives a value of 2.62. Two features stand out: frequency of 'Toys_and_Hobbies' and 'Baby Product'. This suggests that, for this household, making frequent purchases in these categories pushes the model to predict a higher household size. This makes sense as larger household sizes are more likely to have children, and these two purchase categories are commonly associated with families with children. In contrast, spending on products such as grocery & food and kitchen & dining has either very small or slightly negative effects in this case.

For Example 2, the random forest model predicts a household size of about 2.13, while the local approximation model predicts approximately 2.38. Two features stand out with positive contributions: frequency of 'Baby_Product' and 'Total Spent', which increase the predicted household size. On the other hand, the frequency of 'Grocery and Food' shows a small negative contribution in this household's case, lowering the prediction slightly. So this household's prediction is mainly driven by purchases in baby categories and overall spending.

4.1.2 Model 2 (lrn_rf1)

This random forest model used the ranger implementation for regression. To create our final model, we initially performed hyperparameter optimization/tuning to select the combination that gave the best predictive performance on the training data. The hyperparameters we tuned were num.trees, mtry, min.node.size, and sample.fraction. The final model was then retrained on the training split (80% of the data).

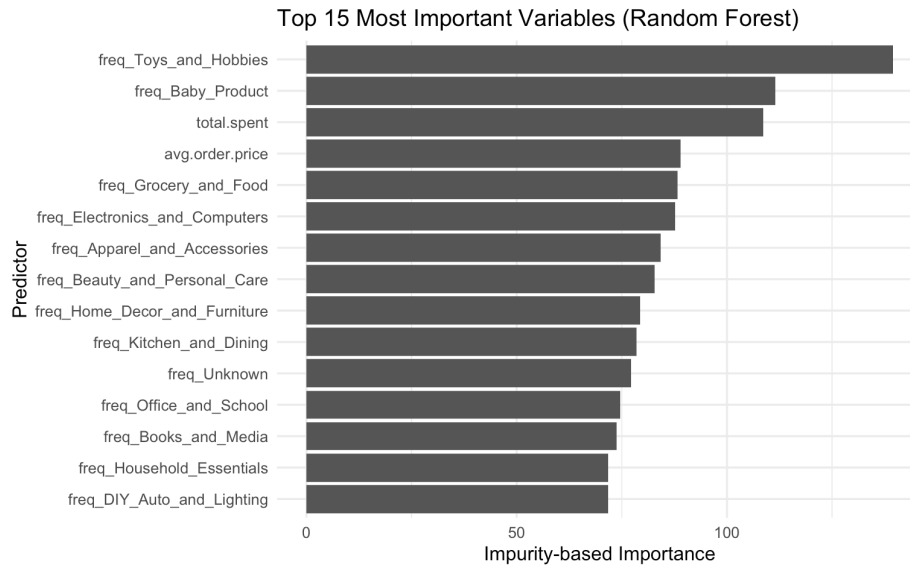


Figure 3: Random Forest Model - Variable Importance Plot

Based on Figure 3, the top three predictors of household size are the frequency of purchases in the ‘Toys and Hobbies’ category, the frequency of ‘Baby Product’ purchases, and the ‘Total Spent’ amount. This is very similar to the results we got from our first model’s (tuned_lrn_xgboost1) variable importance plot.

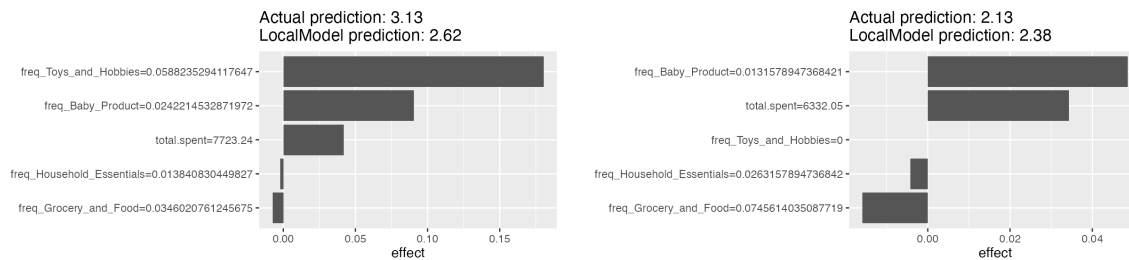


Figure 4: Left – Observation 1, Right – Observation 20

For observation 1, the actual predicted household size from the random forest model is approximately 3.13, while the local surrogate model approximation gives a value of 2.62. Two features stand out: frequency of ‘Toys_and_Hobbies’ and ‘Baby Product’. This suggests that, for this household, making frequent purchases in these categories pushes the model to predict a higher household size. This makes sense as higher household sizes are more likely to have children, and these two purchase categories are commonly associated with families with children. In contrast, spending on products such as ‘Grocery and Food’ and ‘Kitchen and Dining’ has either very small or slightly negative effects in this case.

For Example 2, the random forest model predicts a household size of about 2.13, while the local approximation model predicts approximately 2.38. Two features stand out with positive contributions: frequency of ‘Baby_Product’ and ‘Total Spent’, which increase the predicted household size. On the other hand, the frequency of ‘Grocery and Food’ shows a small negative

contribution in this household's case, lowering the prediction slightly. So this household's prediction is mainly driven by purchases in baby categories and overall spending.

4.1.3 Model 3 (xgb_tune4)

For this third candidate model, we fit an XGBoost, first tuning hyperparameters eta, max_depth, colsample_bytree, subsample, lambda, and alpha with 5-fold CV resampling on the entire training dataset and a random search tuning method with 60 evals, using regr.rmse as the performance measure. We also used the “spend_” features, which we created by multiplying total.spent by each umbrella category frequency. We then trained the final model on 70% of the train set and evaluated its final performance on the 30% train → test split. Below is the resulting importance plot:

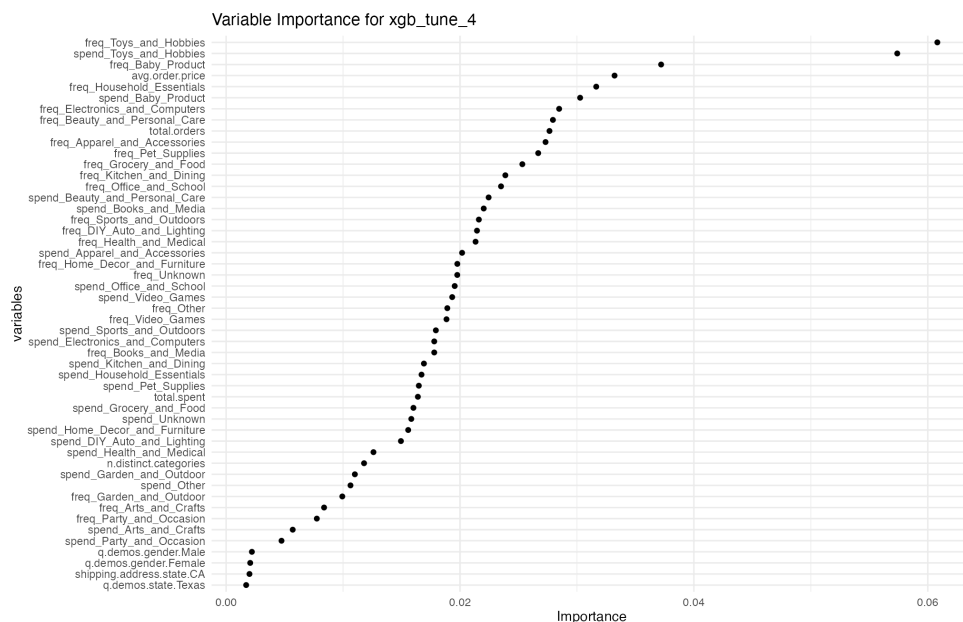


Figure 5: XGBoost Model 2 - Variable Importance Plot

Based on Figure 5, the top three predictors of household size are the frequency of purchases in the ‘Toys and Hobbies’ category, the total spent on the ‘Toys and Hobbies’ category, and the frequency of purchases in the ‘Baby Product’ category. This is very similar to the results we got from our first model’s (tuned_lrn_xgboost1) variable importance plot.

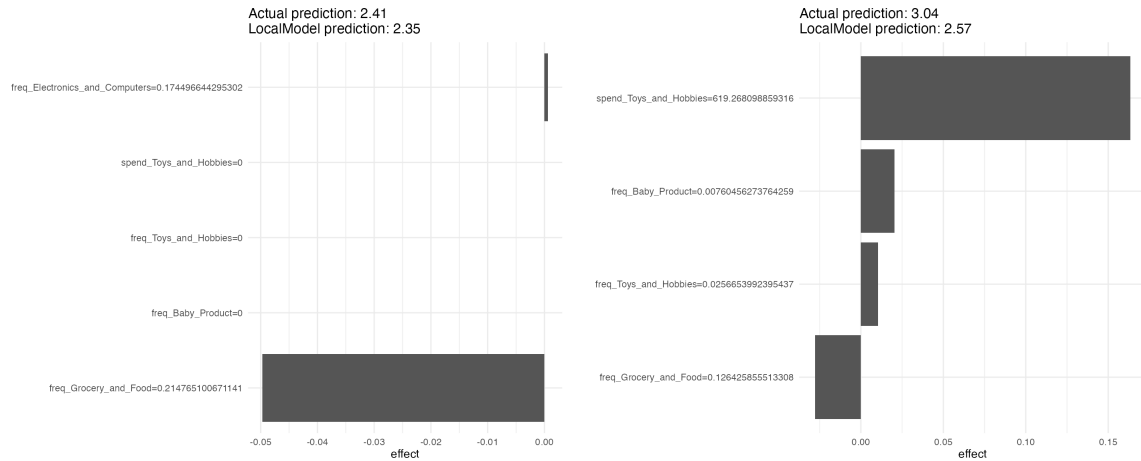


Figure 6: Left – Observation 1, Right – Observation 20

For observation 1, the actual predicted household size from the XGBoost model is approximately 2.41, while the local surrogate model approximation gives a value of 2.35. One feature stands out: ‘freq_Grocery_and_Food.’ For this household, making frequent purchases for grocery and food-related products have a negative effect in this case. In contrast to what we’ve seen in Figures 4 and 2, the frequency of purchasing products in toy and baby-related categories don’t seem to have a strong positive effect.

For Example 2, the XGBoost model predicts a household size of about 3.04, while the local approximation model predicts approximately 2.57. Only one feature stands out with a positive contribution: ‘spend_Toys_and_Hobbies,’ which increases the predicted household size. On the other hand, the frequency of ‘Grocery and Food’ shows a small negative contribution in this household's case, lowering the prediction slightly. So this household’s prediction is mainly driven by purchases in toys and hobbies, which aligns with our previous findings.

4.2 Model Evaluation, Tuning, and Interpretive Graphs

Model identifier	Metric score
tuned_lrn_xgboost1	MSE: 1.170318 RMSE: 1.081812
lrn_rfl	MSE: 1.159199 RMSE: 1.076661
xgb_tune4	MSE: 1.0998414 RMSE: 1.0487332

Table 2: Summarization of the Performance of Each Model

As a preliminary note, all models were trained and validated using 5-fold cross-validation, and model performance was assessed using root mean squared error. The benchmark() function performed a 5-fold cross-validation for all three models and aggregated the results into a

benchmark object. We also compared our models with a featureless baseline model that represents the mean of our target variable within our training data.

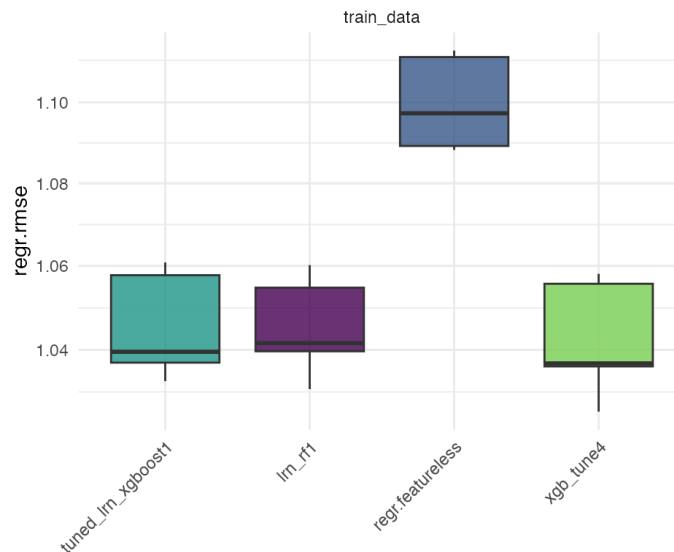


Figure 6: Benchmark Experiment with All Models

After running the benchmark experiment, we found the model with the lowest aggregated RMSE value was our second XGBoost model (xgb_tune4). Our two XGBoost models seemed very close in performance based on what is shown in Figure 6, but the median value was smaller for the second XGBoost model and showed to be more right-skewed, making us believe that this model would have more prediction averages smaller than the first XGBoost model (tuned_lrn_xgboost1). We ultimately decided that xgb_tune4 should be our model as the model with the smallest RMSE makes the most sense to be the best-performing model.

5 Discussion of Final Model

We decided on our second XGBoost model (xgb_tune4) as our final model because it produced the smallest RMSE and MSE values out of all our models. When testing our predictions in Kaggle, our team scored the highest with the values predicted by this model. Additionally, our team expected the XGBoost model to perform the best, as it typically outperforms other models, such as random forest, in Kaggle competitions.

Although our model was tuned to optimize its hyperparameters, the tuning process could have missed the very best hyperparameters, meaning there could be other hyperparameters that can make our current model more optimal. Instead of using a random search tuner, changing the tuner to grid search could have found better hyperparameters. We could also have tuned the subsample ratio of columns for each level of the tree and potentially obtained better predictions this way. While a 5-fold cross-validation is effective, repeated cross-validation would have been an assuring process to conduct to gain more confidence in our model. However, doing this would be computationally expensive and extremely time-consuming. Additional data that could be useful to improve our predictions is to use our total spent and total orders, and determine how

much a household typically buys per order – families tend to have larger grocery proportions, especially households with kids.

6 Appendix

U.S. Census Bureau, Population Division. DP05ACS Demographic and Housing Estimates. (2022)

7 Contributions made by each team member

Main Contributors: Charlie Kopp, Angela Luo, Maureen Widjaja

Editors and Oversight: Eddie He, Trevor Nghiem